Methods

1. Data

We use annual real GDP per capita from the Maddison Project Database (2023 release), PPP-adjusted, for a fixed panel of $N = 26$ economies: Australia, Austria, Belgium, Canada, Denmark, Finland, France, Germany, Greece, India, Indonesia, Italy, Japan, Mexico, the Netherlands, Norway, Portugal, South Africa, South Korea, Spain, Sweden, Switzerland, Thailand, Turkey, the United Kingdom and the United States. The panel is balanced: every economy has an uninterrupted annual observation from 1950 to 2022, so no interpolation, gap-filling or unbalanced-panel weighting enters the pipeline at any stage.

We do not assert a price base year, and none is needed. The pipeline operates on demeaned logs: any multiplicative rebasing constant κ common to a series enters as $\ln(\kappa\,\mathrm{GDPpc}) = \ln\kappa + \ln\mathrm{GDPpc}$ and is removed exactly by the subsequent mean subtraction. Every quantity in this paper — cycles, phases, correlations, order parameter — is therefore invariant to the choice of base year and to PPP rebasing.

Two distinct date ranges are used deliberately and should not be conflated. Filtering and phase extraction are performed on the full 1950–2022 series, so that the band-pass filter and the Hilbert transform have run-in on both sides of the analysis window. All statistics are then computed on the **1970–2022** window ($T = 53$ annual observations), discarding the first two decades. This protects every reported quantity from the filter and Hilbert edge transients, which are severe in short annual records. It is a consequential choice rather than a formality: recomputing phases on a 1970–2022 series alone, with no run-in, changes the order parameter materially in individual years and shifts the synchronization–amplitude coefficient of §6.2 by roughly 0.08 (§7.2).

2. Cycle extraction

For each economy i we take the natural log of real GDP per capita, $y_i(t) = \ln \mathrm{GDPpc}_i(t)$, and remove its sample mean. We isolate business-cycle frequencies with a second-order Butterworth band-pass filter applied by forward–backward filtering (`filtfilt`), which is zero-phase and therefore introduces no artificial lead or lag between economies — a necessary property, since the phase relationships between economies are the object of study.

The pass-band is 2–9 years. In normalized frequency units at an annual sampling rate ($f_{\mathrm{Nyq}} = 0.5\ \mathrm{yr}^{-1}$) the band edges are $f_{\mathrm{low}} = 1/9\ \mathrm{yr}^{-1}$ and $f_{\mathrm{high}} = 0.49\ \mathrm{yr}^{-1}$. The upper edge is set at 0.49 rather than the nominal 0.5 because a Butterworth design requires the critical frequency to lie strictly inside the open interval $(0, f_{\mathrm{Nyq}})$; the resulting cycle band is thus effectively 2.04–9 years. The band brackets the conventional business-cycle range and is not tuned to any outcome; §7.1 reports the sensitivity of every result to the band edges.

Write the filtered cycle for economy i as $c_i(t)$. We form the analytic signal by Hilbert transform,

$$z_i(t) = c_i(t) + \mathrm{i}\,\mathcal{H}[c_i](t) = A_i(t)\,e^{\mathrm{i}\theta_i(t)},$$

which yields an instantaneous phase $\theta_i(t) = \arg z_i(t)$ and an instantaneous amplitude $A_i(t) = |z_i(t)|$ for each economy and year.

3. Synchronization measure

Treating the 26 economies as a population of phase oscillators, we measure collective synchronization by the Kuramoto order parameter

$$r(t)\, e^{\mathrm{i}\psi(t)} = \frac{1}{N} \sum_{i=1}^{N} e^{\mathrm{i}\theta_i(t)},$$

where $r(t) \in [0, 1]$ is the modulus of the mean phase vector and $\psi(t)$ is the collective phase. $r = 0$ corresponds to uniformly dispersed phases and $r = 1$ to identical phases. We also define the mean cycle amplitude $\bar{A}(t) = N^{-1} \sum_i A_i(t)$, a measure of how large the swings are, independent of whether they are aligned. Throughout, $A_i(t)$ denotes a single economy's instantaneous amplitude and $\bar{A}(t)$ the cross-sectional mean; no other amplitude symbol is used.

Because $r(t)$ is a modulus, it is non-negative and biased away from zero in finite samples; with $N = 26$ the expected value under independent uniform phases is $\approx N^{-1/2} \approx 0.20$. We therefore never interpret the level of r against zero, only its variation over time and against matched surrogates (§5).

4. Crisis dating

Crisis years are taken exogenously from the Kose–Terrones definition of a **global recession**, which identifies contractions in world real GDP per capita accompanied by broad-based declines in other measures of global activity. This yields five dates in our window: **1975, 1982, 1991, 2009 and 2020**. These dates are fixed before any statistic is computed and are never adjusted, re-weighted, or selected on the basis of the synchronization series. We use only the crisis *onset* years; no crisis is assigned a duration.

The count matters for interpretation and is stated plainly: five events in 53 years. Every crisis-conditioned test in this paper therefore rests on five degrees of freedom at most, and the leave-one-out analysis of §7.3 shows that the effective count for the pre-crisis result is smaller still.

5. Null models

Two nulls are used, for two different purposes.

5.1 Circular-shift permutation. For any test that correlates a continuous series with the crisis indicator, we build the null by circularly shifting the crisis indicator vector G — a binary vector with ones at the five crisis years — through all $T - 1 = 52$ non-identity rotations, recomputing the statistic each time (53 configurations in total, counting the unshifted one). This preserves both the autocorrelation of the continuous series and the number and relative spacing of the crisis dates, and destroys only their alignment with the series. The p-value is the fraction of shifts whose statistic equals or exceeds the observed one, with the usual +1 correction in numerator and denominator.

This null has a hard resolution limit that must be quoted alongside any p it produces: with 52 shifts, the smallest attainable p-value is $1/53 = 0.019$. A reported $p = 0.019$ therefore means "no shift beat the observed value" and cannot be read as evidence of a smaller effect size than $p = 0.019$; it is a floor, not a measurement.

5.2 Spectrum-preserving surrogate economy sets. For tests concerning the structure of the oscillator population, we generate 5,000 surrogate 26-economy sets by Fourier phase randomization. We take the real FFT of each economy's band-passed cycle, retain the amplitude spectrum exactly, replace each non-trivial Fourier phase with an independent draw from $U(0, 2\pi)$ (holding the DC and, for even record length, Nyquist phases real), and invert the transform. Each surrogate economy therefore has the same power spectrum, and hence the same autocorrelation and cycle periodicity, as its real counterpart, but its phase relationship to every other economy is destroyed. Phases are randomized independently across economies, so the surrogate ensemble is the null of *spectrally identical but mutually unsynchronized* economies. Surrogate p-values with 5,000 draws have a floor of $1/5001 = 2 \times 10^{-4}$.

6. Tests

6.1 Loss-of-diversification decomposition. The world output cycle is the weighted aggregate of the 26 band-passed economy cycles,

$$C_w(t) = \sum_{i=1}^{N} w_i \, c_i(t),$$

with weights $w_i \geq 0$, $\sum_i w_i = 1$. On each rolling 10-year window its variance decomposes exactly into an own-term and a cross-term,

$$\mathrm{Var}[C_w] \;=\; \underbrace{\sum_i w_i^2 \, \sigma_i^2}_{\text{own}} \;+\; \underbrace{\sum_{i \neq j} w_i w_j \, \sigma_{ij}}_{\text{cross}},$$

where σ_i^2 is economy i's within-window cycle variance and σ_{ij} the pairwise covariance. Variances are computed with the population convention (dividing by the window length $W = 10$). The own-term is what world output would fluctuate by if the 26 economies were uncorrelated; the cross-term is the co-movement surcharge. The decomposition is an identity, not a model, and holds window by window.

The test is deliberately *not* "does world variance rise with average correlation." That is algebraically guaranteed: for equal weights the cross share is cross/total $= (N-1)\bar{\rho}/(1+(N-1)\bar{\rho})$, so with $N = 26$ the 325 covariance pairs outnumber the 26 variances by more than an order of magnitude and the cross-term dominates at any $\bar{\rho}$ above a few hundredths. Observed $\bar{\rho}$ ranges from 0.13 to 0.88 across our windows, which alone forces a cross share between 0.76 and 0.96. Reporting a high cross share as a finding would therefore be reporting arithmetic. The non-circular question is which term supplies the *excess* variance in global-recession years relative to normal years:

$$\Delta_{\text{own}} = \langle \text{own} \rangle_{\text{crisis}} - \langle \text{own} \rangle_{\text{normal}}, \qquad \Delta_{\text{cross}} = \langle \text{cross} \rangle_{\text{crisis}} - \langle \text{cross} \rangle_{\text{normal}},$$

summarized as the cross-term's share of the excess, $\Delta_{\text{cross}}/(\Delta_{\text{own}} + \Delta_{\text{cross}})$.

Two weightings are computed and both are reported. **GDP-weighted** (w_i proportional to the economy's within-window mean real GDP, i.e. GDP per capita times population) is the primary framing, since it is the actual world economy. **Equal-weighted** ($w_i = 1/N$) is the robustness check and ties directly to the equal-weight order parameter of §3, which gives every economy one vote. Because the 10-year window is taken entirely inside the analysis window, the decomposition is defined from 1979 onward, leaving four of the five global recessions (1982, 1991, 2009, 2020) inside its support; 1975 falls before the first complete window and does not enter this test. This is a material reduction in an already small event count and is carried into the interpretation.

Across 1979–2022 the cross-term supplies the great majority of the excess crisis variance under both weightings: 82.4% GDP-weighted and 96.9% equal-weighted. The underlying cross-shares are 0.723 in crisis windows against 0.669 in normal windows (GDP-weighted), and 0.880 against 0.818 (equal-weighted). The direction is consistent across weightings and the gap is in the third decimal place of the share — a shift of five to six percentage points on a base already above two-thirds, exactly as the algebraic baseline above would lead one to expect. The equal-weighted figure is the larger because equal weights give the 325 covariance pairs their maximum leverage; the GDP-weighted figure, in which a few large economies carry most of the own-term, is the one we treat as describing the world economy.

Significance is assessed against the spectrum-preserving surrogate ensemble of §5.2. §7.6 reports the outcome of that test, which does not behave as a well-posed significance test for the share statistic, and states what we do and do not claim as a result.

6.2 Synchronization–amplitude coupling. We correlate $r(t)$ with $\bar{A}(t)$ over 1970–2022 by Pearson correlation, in two versions: on the raw series, and after removing a slow trend from each by subtracting a Gaussian-smoothed copy ($\sigma = 8$ yr). The detrended version is the headline: both series carry secular drift over five decades, and the raw correlation cannot distinguish co-movement at business-cycle timescales from two coincident long-run trends. The two versions agree closely — +0.530 raw against +0.608 detrended, at the same $p = 0.019$ — so the result does not hinge on the detrending choice; we report the detrended value because it isolates business-cycle timescales, not because it is the larger. Significance is assessed by the circular-shift null of §5.1 applied to the paired series, and separately against the surrogate ensemble of §5.2. This test uses no crisis dates at all.

6.3 Crisis-relative lead–lag profile. For each lag $\ell \in \{+2, +1, 0, -1, -2\}$ we correlate $r(t)$ with the crisis indicator shifted by ℓ, so that positive ℓ tests whether synchronization *precedes* crises and negative ℓ whether it falls after them. Leading lags are tested one-sided upward and trailing lags one-sided downward, matching the pre-registered direction of the hypothesis; the contemporaneous lag is tested upward. Significance uses §5.1.

Five lags are tested, and the raw p-values are reported as such alongside a multiplicity correction over the five (§7.4). We report the raw values because they are the natural output of the permutation scheme, and the corrected values because five lags were in fact examined.

6.4 Composite crisis epoch. We average r and $\bar{A}$ over a ±4-year window centred on each of the five crisis onsets and take the mean across events, giving a composite

epoch profile for each quantity. The peak lag of each profile is read off directly. This is a descriptive alignment, not a test, and is reported with no p-value.

6.5 Network geometry. On 10-year rolling windows we compute the 26×26 Pearson correlation matrix of the band-passed cycles, convert it to the standard ultrametric-compatible distance $d_{ij} = \sqrt{2(1-\rho_{ij})}$, build the minimum spanning tree by Prim's algorithm, and record its mean edge length $L(t) = (N-1)^{-1} \sum_{e \in \mathrm{MST}} d_e$. Falling L indicates a tightening network. Because the 10-year window needs nine years of run-in, $L(t)$ is defined from 1971 onward. Two questions are asked of L: whether it trends secularly — reported as a description via decade means (0.706 in the 1970s against 0.379 in 2013–2022, a factor of 1.9×; the single-year endpoints 0.771 and 0.250 are noisy individual windows and are not the interpreted quantity) — and whether contraction coincides with the pre-crisis windows. The latter is quantified by a Kendall-τ trend over the 8 years preceding each onset, averaged over crises, and tested one-sided downward against the surrogate ensemble of §5.2, both on raw L and on Gaussian-detrended L. Both are null: $\tau = -0.171$, $p = 0.167$ on raw L and $\tau = -0.086$, $p = 0.316$ detrended. The test is described here so that it is on record as having been run and failed, not inferred from its absence.

6.6 Phase-lag structure. For each economy we compute its mean phase offset from the collective phase, $\langle \arg e^{\mathrm{i}(\theta_i - \psi)} \rangle$, and convert it to years by dividing by the mean angular frequency $\bar{\omega}$ (estimated as the mean year-on-year phase increment across economies, giving a mean cycle period of 5.85 years). An economy with a consistently positive offset sits ahead of the collective phase; whether that constitutes economic leadership is a question for Paper 2 and is not addressed here. The test statistic is the standard deviation of these 26 lags, observed at 0.403 years: a large spread would leave leader–follower structure recoverable, a small spread implies the economies move near-simultaneously. It is compared to the surrogate ensemble of §5.2. Because the scientific question is whether the observed spread is *smaller* than chance, the test is one-sided **downward**: the observed 0.403 yr lies below all 5,000 surrogate values (surrogate mean 0.942, minimum 0.446), giving $p = 2 \times 10^{-4}$, the ensemble resolution floor. The economies are phase-locked tightly enough that no epicenter is recoverable.

6.7 Critical slowing down. We test the standard early-warning battery — rising lag-1 autocorrelation, rising variance, and rising restoring-time — on the pre-crisis windows. All three are reported as null in Results. Annual data over 8-year pre-crisis windows gives these estimators very few effective degrees of freedom, so the null is weak evidence in the statistical sense and we do not read it as establishing an absence.

It is not, however, an empty result. Rising autocorrelation and variance are the specific signature of a system approaching a *bifurcation* — the slow eigenvalue going to zero as a fixed point loses stability. In the Ashwin–Wieczorek–Vitolo–Cox (2012) taxonomy this is B-tipping, and it is only one of three routes; noise-induced (N-) and rate-induced (R-) tipping produce no such precursor, because in neither case does the system's restoring rate vanish before the transition. The absence of the slowing-down fingerprint is therefore consistent with a non-bifurcation transition, and is informative about *which class* of tipping is in play even where it is uninformative about proximity to a threshold. We record the classification here and leave the mechanism argument to the Discussion; nothing in this paper identifies which non-bifurcation route applies.

7. Robustness and disclosure

Every check in this section was run on the final pipeline, and all outcomes are reported whether or not they support the paper's claims.

7.1 Filter band. Results are recomputed for pass-bands of 2–8 and 3–10 years. The pre-crisis lead is present in all three bands but its p-value moves across the conventional threshold: lead-1 gives $p = 0.019$ (2–9 yr), 0.057 (2–8 yr) and 0.038 (3–10 yr). The post-crisis decline is not robust to the band: the lag -2 coefficient is -0.28 ($p = 0.019$) at 2–9 years but only -0.02 ($p = 0.45$) at 3–10 years.

7.2 Detrending method. Substituting an HP filter ($\lambda = 100$) for the band-pass leaves the lead result materially unchanged (lead-1 +0.27, $p = 0.019$). Using first differences of log GDP per capita instead weakens the lead-2 term to $p = 0.094$ and eliminates the post-crisis decline entirely (-0.01, $p = 0.51$). The post-crisis result is thus method-dependent in a way the pre-crisis result is not. Separately, as noted in §1, the run-in convention moves this coefficient materially: computing both r and $\bar{A}$ on the 1970–2022 window without run-in gives +0.531 detrended against +0.608 with run-in, at the same $p = 0.019$. We report the run-in-protected value throughout, on the ground that the edge transients it removes are an artifact, not because it is the larger; the sign and significance of the result are the same either way.

7.3 Leave-one-crisis-out. Dropping each crisis in turn and refitting §6.3 shows that the pre-crisis lead is not evenly carried by the five events. It survives removal of 1975, 1982 and 1991, but dropping either 2009 or 2020 destroys the lead-1 significance, and dropping both leaves no lead at all. A split-sample fit tells the same story: 1970–1995 (crises 1975, 1982, 1991) shows no significant lead, while 1996–2022 (crises 2009, 2020) does. The effective event count behind the pre-crisis claim is therefore $n \approx 2$, both in the modern era, and the claim is stated in the paper as an era-specific property of the tightly-coupled economy rather than a fifty-year regularity.

7.4 Multiplicity. Across the five lags of §6.3, no coefficient survives a Bonferroni correction (lead-1 and lag -2: $p_{\text{adj}} = 0.095$; lead-2: 0.189). Under Benjamini–Hochberg control of the false discovery rate, lead-1 and lag -2 reach $q = 0.047$ and lead-2 $q = 0.063$. The permutation floor of §5.1 caps how much correction the raw values can absorb: 0.019 is already the smallest value the test can return.

7.5 Leave-one-economy-out. Removing the United States, Germany or Japan individually leaves the lead–lag profile essentially unchanged (lead-1 $p = 0.038$, 0.019, 0.019 respectively). No single large economy drives the result.

7.6 Surrogate test of the decomposition. The share statistic of §6.1 does not admit a well-behaved surrogate test, and we report this rather than substituting a statistic that does. Its denominator $\Delta_{\text{own}} + \Delta_{\text{cross}}$ is a difference of window means that is near zero and of either sign under the surrogate null, so the ratio is unbounded and its null distribution is heavy-tailed and non-monotone in the quantity of interest. Against 5,000 surrogate sets, 2,385 draws exceed the observed equal-weighted share and 1,741 exceed the GDP-weighted share — nominal p-values of 0.48 and 0.35, which reflect the pathology of the ratio rather than the strength of the effect. Tested on the unnormalized components instead, both Δ_{cross} and Δ_{own} exceed every one of the 5,000 surrogates ($p = 2 \times 10^{-4}$, the ensemble resolution floor), but that test is uninformative for our

purpose: it rejects the null that crisis windows resemble phase-randomized ones at all, which is a much weaker statement than the decomposition claim, and it rejects for the own-term too.

We therefore state the decomposition result as a descriptive magnitude with an explicit floor, not as a significance claim: the cross-term supplies the large majority of the excess crisis variance under both weightings, on four events, and the algebraic baseline of §6.1 already guarantees a high cross share before any economics enters. What the data add beyond arithmetic is the *gap* between crisis and normal windows (§6.1), which is modest in absolute terms. A defensible significance test for this statistic — one whose null distribution is proper — is not available to us and we do not report one.

7.7 Window-convention sensitivity of the network measure. $L(t)$ is reported on 10-year windows, and its early values depend on whether the window is allowed to reach back into the 1950–69 run-in. Restricting windows to 1970 onward, L is defined from 1979 and the decade-mean comparison becomes 0.652 (1979–89) against 0.379 (2013–2022). The convention is stated wherever L is quoted.

8. Reproducibility

The input data are the Maddison Project Database (2023 release) as distributed in the `MaddisonData` R package (v1.1.0). The pipeline is deterministic apart from the surrogate ensembles, which are seeded. Rebuilt from the raw database, it reproduces the order-parameter series to floating-point round-off (3×10^{-16}) and reproduces every lead–lag coefficient and p-value exactly. The equal-weighted variance decomposition of §6.1 reproduces to 1.5×10^{-18}.

Two quantities in earlier internal result files could not be reproduced from the raw data and are reported here at their rebuilt values: the synchronization–amplitude correlation (§6.2, rebuilt +0.608 detrended) and the correlation between mean pairwise cycle correlation and the order parameter (rebuilt +0.45 to +0.51 depending on the run-in convention). We were unable to recover the exact amplitude convention behind the earlier figures; the qualitative conclusions are unaffected in both cases, and the rebuilt values are the ones stated throughout.

Figure — Methods diagnostics. Panel **a**: MST mean edge length by year (grey) with decade means (blue) and the single-year endpoints (red) that produce the ∼ 3× ratio; decade means give 1.9× and are the interpreted quantity. Panel **b**: the phase-lag spread against its 5,000-member surrogate null — the observed 0.403 yr lies below every surrogate, a lower-tail result at the ensemble floor. Panel **c**: raw and Bonferroni-corrected p-values for the five lags of §6.3, against the 1/53 permutation floor (dotted) and 0.05 (dashed).